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GNN-SDM: Species Distribution Modelling with Graph Neural Networks for Swiss Flora at 30 m Resolution

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Abstract

Traditional species distribution models treat occurrences as independent points, ignoring the spatial structure of landscapes that governs habitat connectivity and species dispersal. We adapt the GNN-SDM framework (**Wu2025**) to Swiss vascular plants at 30 m resolution, constructing a landscape graph of 166,464 patches linked by 465,833 adjacency edges. Using a GraphSAGE architecture tuned for flora, we model habitat suitability for 3,751 species and compare against a Random Forest baseline. [Results summary to be added.]

Contents

1 Introduction

TODO: Write introduction.

2 Data and Study Area

2.1 Study Area

The study area encompasses the entirety of Switzerland and a surrounding buffer zone, defined by the spatial extent of the CHELSAch high-resolution climate dataset (**CHELSAch**). The bounding box spans from 45.57° N to 48.07° N latitude and 5.72° E to 10.66° E longitude, covering approximately 41,400 km^2 of land area. This extent includes a border buffer of roughly 10 km into neighbouring countries to avoid edge effects in spatial analyses such as flow accumulation and neighbourhood-based feature computation.

All environmental layers are aligned to a common 30 m master grid derived from the Copernicus GLO-30 Digital Elevation Model (**CopernicusDEM**), projected in geographic coordinates (EPSG:4326). Switzerland's topography ranges from 193 m (Lake Maggiore) to 4,634 m (Dufourspitze), creating steep environmental gradients over short distances—a property that motivates the use of high-resolution modelling.

2.2 Environmental Features

We compiled 21 candidate environmental features from four thematic groups (Table 1): terrain, land cover, climate, and biological/anthropogenic variables. All layers were processed to the 30 m master grid using source-appropriate resampling methods.

Table 1: Environmental predictor variables compiled for the study. All layers are aligned to the 30 m master grid. The 12 features selected for modelling (after feature selection, Section 2.5) are marked with •.

Group	**Feature**	**Source**	**Native res.**	**Sel.**
Terrain	Elevation	Copernicus GLO-30 DEM	30 m	•
	Slope	Derived from DEM	30 m	•
	Aspect (sin, cos)	Derived from DEM	30 m	•
	Topographic Wetness Index	D8 flow accumulation	30 m	•
	Profile curvature	2^{nd}-order finite diff.	30 m	
	Plan curvature	2^{nd}-order finite diff.	30 m	
Land cover	Tree cover fraction	ESA WorldCover 2021	10 m	•
	Grassland fraction	ESA WorldCover 2021	10 m	•
	Cropland fraction	ESA WorldCover 2021	10 m	•
	Built-up fraction	ESA WorldCover 2021	10 m	
	Snow/ice fraction	ESA WorldCover 2021	10 m	
	Water fraction	ESA WorldCover 2021	10 m	
Climate	Bio1 (annual mean temp.)	CHELSAch	25 m	•
	Bio4 (temp. seasonality)	CHELSAch	25 m	•
	Bio12 (annual precip.)	CHELSAch	25 m	•
	Bio15 (precip. seasonality)	CHELSAch	25 m	
	Bio18 (precip. warmest quarter)	CHELSAch	25 m	
Biological	Canopy height	ETH Global Canopy Height	10 m	•
	NDVI (annual max)	GIMMS MODIS	250 m	
	Human Footprint	**Mu2022**	1 km	•
	Forest-edge distance	Derived from tree cover	30 m	

Terrain features. Elevation was obtained directly from the Copernicus GLO-30 DEM (**CopernicusDEM**). Slope and aspect were derived using first-order finite differences at 30 m resolution. Aspect was encoded as two variables (sin and cos of the azimuth angle) to avoid the circular discontinuity at 0°/360°. The Topographic Wetness Index (TWI) was computed using D8 flow direction and accumulation via the `pysheds` library, with pit-filling and depression resolution applied beforehand. Profile and plan curvature were derived from second-order finite differences and smoothed with a 5×5 mean filter to reduce noise.

Land cover. Six land-cover fraction layers were computed from ESA WorldCover 2021 (**ESAWorldcover**) at its native 10 m resolution. For each class, a binary mask was reprojected to the 30 m grid using average resampling, yielding the fraction of each 30 m pixel covered by that land-cover type. The six classes retained for Switzerland are: tree cover, grassland, cropland, built-up, snow/ice, and permanent water bodies.

Climate. Five bioclimatic variables were derived from CHELSAch monthly temperature and precipitation normals (1981–2010) at 25 m resolution (**CHELSAch**): annual mean temperature (Bio1), temperature seasonality (Bio4, standard deviation), annual precipitation (Bio12), precipitation seasonality (Bio15, coefficient of variation), and precipitation of the warmest quarter (Bio18). Each variable was regridded to the 30 m master grid using bilinear interpolation.

Biological and anthropogenic features. Canopy height was obtained from the ETH Global Canopy Height Model at 10 m resolution (**Lang2023**) and regridded to 30 m via bilinear resampling. The annual maximum NDVI was computed from GIMMS MODIS 8-day composites (250 m, growing season DOY 97–273) and regridded to 30 m. The global Human Footprint index (**Mu2022**) at 1 km was bilinearly downscaled. Forest-edge distance was computed as a signed Euclidean distance transform from the tree-cover fraction layer (threshold 0.5): positive values indicate distance from forest, negative values indicate depth within forest.

2.3 Species Occurrence Data

Species occurrence records for vascular plants in Switzerland were obtained from the Global Biodiversity Information Facility (GBIF) open data snapshot hosted on Amazon S3 (**GBIF2024**). The raw dataset was filtered to retain only records within the study extent that met the following quality criteria: coordinate uncertainty ≤ 1000 m, valid geographic coordinates, and taxonomic rank at species level.

After filtering, the dataset comprised approximately 14 million occurrence records spanning 7,912 vascular plant species. For modelling purposes, we retained only species with at least 100 unique occurrence records, yielding 3,756 qualifying species. This threshold ensures sufficient training data for both the Random Forest baseline and the GNN-SDM, while acknowledging that it excludes many rare and threatened species with sparse observational records.

2.4 Protected Areas

Protected area boundaries for Switzerland were obtained from the World Database on Protected Areas (WDPA), filtered to nationally designated sites within the study extent. The dataset includes national parks, nature reserves, landscape protection areas, and Ramsar sites. These boundaries are used in the gap analysis (**??**) to assess the degree to which high-priority habitat identified by the GNN-SDM falls within existing protected areas.

2.5 Feature Selection

To reduce multicollinearity and computational cost, we performed feature selection using Random Forest importance scores. A patch-level RF classifier was trained for six representative species spanning different habitat types and prevalence levels. Feature importance was computed as the mean decrease in Gini impurity, averaged across species.

From the initial 21 candidate features, 12 were retained for downstream modelling (Table 1). The excluded features were either highly correlated with retained variables (e.g., Bio15 with Bio12; plan curvature with profile curvature) or contributed negligible predictive power (e.g., NDVI at 250 m resolution added little beyond the 10 m canopy height and tree-cover fraction). The selected feature set balances ecological interpretability with predictive performance while keeping the dimensionality manageable for the Self-Organising Map clustering step.

3 Methods

3.1 Landscape Patch Construction

3.1.1 SOM Clustering

TODO: SOM methodology.

3.1.2 Connected Components and Patch Merging

TODO: Patch construction.

3.1.3 Patch Network

TODO: Network construction.

3.2 Baseline: Random Forest

TODO: RF methodology.

3.3 GNN-SDM

3.3.1 Model Architecture

TODO: GNN architecture.

3.3.2 Training Procedure

TODO: Training details.

4 Results

4.1 Random Forest Baseline

TODO: RF results.

4.2 GNN-SDM Performance

TODO: GNN results and comparison.

4.3 Common vs. Vulnerable Species

TODO: Species-level analysis.

4.4 Habitat Suitability Maps

TODO: Map figures.

5 Discussion

5.1 Architecture Adaptation for Flora

TODO: Architecture discussion.

5.2 Value of Graph Structure

TODO: Graph structure value.

5.3 Limitations

TODO: Limitations.

5.4 Future Work

TODO: Future directions.

6 Conclusion

TODO: Write conclusion.

Acknowledgements

TODO: Acknowledgements.